

Note from youtube:

Application layer: It enables the user to access the network resources.

application layer services:

- File transfer and access management (FTAM)
- Mail services
- Directory services

Presentation layer:

It is concerned with the syntax and semantic of the information exchanged between 2 systems

Presentation layer services:

- Translation → converting data common form
- Encryption → protect data
- Compression → reducing number of complex - - [situation]

Session layer:

It establishes, maintains and synchronizes the interaction among communicating devices

Services:

- Dialog control
- Synchronization

Transport layer:

It is responsible for process to process delivery of the entire message.

Services:

- Port addressing
- Segmentation and Reassembly
- Connection control,
- End-to-end flow control
- Error control

Network layer:

It is responsible for delivery data from the original source to the destination network

- logical addressing
- routing

Data link layer:

It is responsible for moving data [frames] from one node another node.

- Framing
- Physical addressing
- Error control
- Access control
- Flow control

Physical layer:

It is responsible for transmitting bits over a medium. It also provides electrical and mechanical specifications

- Physical characteristic of media
- Representation of bits
- Data rate
- Synchronization of bits
- Line configuration
- Physical topology